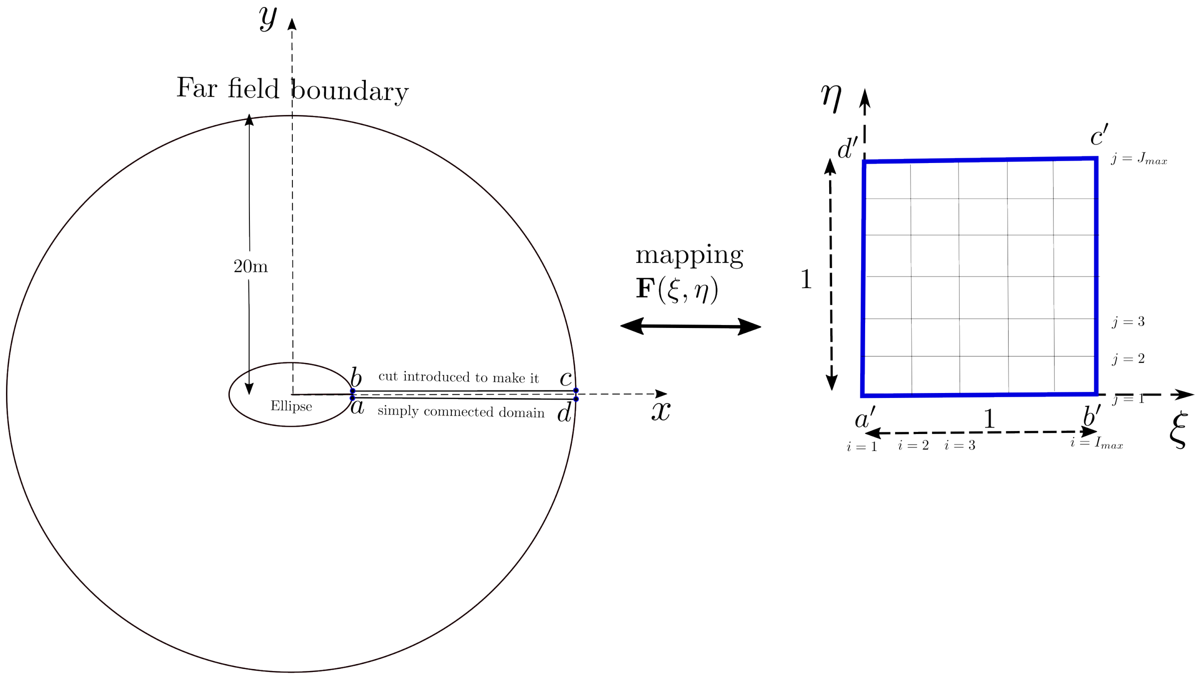
Assignment-2: Grid generation using Transfinite interpolation (TFI) Method

Problem Statement

Consider a physical domain between a ellipse (with semi-major axis a 1 m and semi-minor axis 0.5 m) and far field boundary located at 20m as shown in the Figure 1. You need to derive Transfinite interpolation mapping for generating grid in the physical domain. A (imaginary) cut is introduced so that physical domain becomes simply connected domain, which can then be mapped into unit square in the computational domain as shown in the Figure 1. Note that the co-ordinates of the points a and d are identical to the co-ordinates of b and c respectively.

(1) Derive the transfinite interpolation function/mapping for generating grid inside the physical domain showing every important steps.

(2) Write a C/C++/python code to generate the grid points in the physical domain as specified in the next section.



(i) Physical Domain (ii) Computational Domain

Figure 1: Grid mapping used for circular cylinder

Domain Specification and Discretization

Consider a ellipse (with semi-major axis a 1 m and semi-minor axis 0.5 m) over which incompressible irrotational flow is to be computed. The far field boundary is located at 20 m distance away as shown in the Figure 1.

Generate a grid of the size 101 $\times$ 81, where there are 101 grid points in the ξ-diection and 81 grid points in the η-diection. There will be similarly 101 $\times$ 81 grids in the physical domain which can be obtained using Transfinite Interpolation method.

Report

The report must include the following:

1. Every step of derivation involving the Transfinite interpolation mapping for generating grid in the physical domain as show in the Figure 1.

2. Figure showing the final grid generated by you.

General Instructions

- **Checklist for submission:**
  1. Code for grid generation based on the mapping shown in Figure 1.
  2. A "README.txt" file which contains the proper description on how to run your code and get the plots.
  3. Report, in pdf format, where all the above mentioned derivation and plots are provided.
- **Instruction for submission:**
  - Rename your program file as your roll number (example: 204010006.c).
  - Rename the report as your roll number (example: 204010006.pdf)
  - Upload in MOODLE all the **3 documents** separately as stated in the Checklist above.
  - Don't submit zip file.
- **Notes:**
  - Marks will be given only if the derivation is correct and the program is working and showing correct result.
  - Assignment will not be evaluated if "instruction for submission" are not followed properly.
- **Copying programs from one another or any other source will result in severe penalties. Submissions will be checked using Stanford University's plagiarism detection software, MOSS, which can identify plagiarism even if variable names are altered or program lines are rearranged.**

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